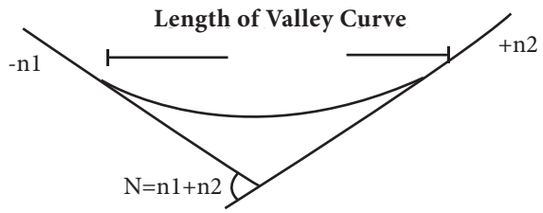


Sumit Curve



Valley Curve

6.4 Highway Materials

The materials required for the construction of highways can be broadly divided into the following two categories.

1. Aggregates
2. Binding materials. (Example: Bituminous materials, cement)

6.4.1 Types of Road Aggregate

The road aggregates can be classified in the following categories.

i. Crushed rock aggregates
ii. Gravel
iii. Sand
iv. Blast furnace slag

Activity 1

Collect the pictures of various types of road materials and prepare an album.

6.4.2 Requirement of a good road aggregate

Following are the desirable properties or requirements of a good road aggregate.

i. Strength
ii. Hardness
iii. Toughness
iv. Durability
v. Shape
vi. Adhesion with bitumen

6.4.3 Test For Road Materials

i) List of tests on sub grade soil

i. Shear tests
ii. Bearing tests
iii. Penetration tests

ii) List of tests on road aggregates

i. Water absorption test
ii. Aggregate crushing test
iii. Aggregate impact test
iv. Attrition test

iii) List of tests in Bituminous material

i. Ductility test
ii. Loss on heat test
iii. Float test
iv. Penetration test
v. Softening point test
vi. Solubility test
vii. Specific gravity test

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